

EXEPERIMENT -10

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Create a data visualization (e.g., pie charts, bar graphs) for an inventory management system using javascript

AIM

The aim is to create data visualizations such as pie charts and bar graphs for an inventory management system using JavaScript.

PROCEDURE

Step 1: Set Up Your HTML File

Create an HTML file to include the canvas elements for charts and link Chart.js.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Inventory Management System</title>
```

```
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
```

```
</head>
```

```
<body>
```

```
<h2>Inventory Management System</h2>
```

```
<canvas id="pieChart" width="400" height="400"></canvas>
```

```
<canvas id="barChart" width="400" height="400"></canvas>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

Step 2: Create the JavaScript File for Charts

Create a JavaScript file to define the data and generate charts.

```
// script.js
```

```
// Data for the inventory
```

```
const inventoryData = {  
  labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],  
  datasets: [  
    {  
      label: 'Items in Stock',  
      data: [200, 150, 100, 80, 50],  
      backgroundColor: [  
        '#FF6384',  
        '#36A2EB',  
        '#FFCE56',  
        '#4BC0C0',  
        '#9966FF'  
      ],  
    },  
  ],  
}
```

```
}  
]  
};
```

```
// Creating the Pie Chart
```

```
const ctxPie = document.getElementById('pieChart').getContext('2d');  
const pieChart = new Chart(ctxPie, {  
  type: 'pie',  
  data: inventoryData,  
  options: {  
    responsive: true,  
    plugins: {  
      title: {  
        display: true,  
        text: 'Inventory Distribution'  
      }  
    }  
  }  
});
```

```
// Creating the Bar Chart
```

```
const ctxBar = document.getElementById('barChart').getContext('2d');  
const barChart = new Chart(ctxBar, {  
  type: 'bar',  
  data: inventoryData,
```

```
options: {  
  responsive: true,  
  plugins: {  
    title: {  
      display: true,  
      text: 'Items in Stock by Category'  
    }  
  },  
  scales: {  
    y: {  
      beginAtZero: true  
    }  
  }  
}  
});
```



